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PRACTICE Series

NEET UG 2027

Physics High-Scoring Concepts, Chapters, Mock Tests & Preparation Guide

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About the eBook

Preparing for the NEET exam requires a strategic approach, especially when it comes to mastering physics. This eBook is your complete guide to excelling in the physics section of NEET. It has been carefully designed to provide you with:

Comprehensive Study Material: Covering every chapter and concept in the NEET syllabus for physics, this eBook ensures a detailed understanding of fundamental and advanced topics.

Chapter-Wise Analysis: Each chapter is broken down into bite-sized, easy-to-digest segments, focusing on conceptual clarity and application-based learning.

Important Topics Highlighted: Stay focused with a list of critical topics frequently asked in past NEET exams, along with tips to prioritise them during your preparation.

Practice Questions: Test your understanding with a wide variety of MCQs and numericals designed to simulate NEET-level difficulty.

Quick Revision Notes: Perfect for last-minute preparation, these notes summarise key formulas and concepts for rapid revision.

Insights from Previous Years' Papers: Get an edge by analysing trends and types of questions asked in previous NEET exams.

Whether you're revising or starting afresh, this eBook is tailored to guide you every step of the way to a top score in the physics section.

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A Note to the Students

Dear Students,

The journey to cracking NEET is as challenging as it is rewarding. As you immerse yourself in preparation for the NEET exam, remember that consistency, focus, and a positive mindset are your greatest allies.

- **Plan Wisely:** Stick to a well-structured schedule that balances studying, practice, and self-care.
- **Master the Basics:** Understanding core formulas and derivations is crucial for excelling in physics. Don’t hesitate to revisit foundational concepts.
- **Practice, Practice, Practice:** Solve as many numericals as you can. It strengthens your grasp of the subject and boosts confidence.
- **Use Resources Wisely:** This eBook is your tool to navigate physics effectively – make the most of it.
- **Stay Motivated:** Challenges are inevitable, but persistence will lead you to success. Surround yourself with positivity and believe in your potential.

We wish you the very best for your preparation and are confident that with dedication and hard work, success will be yours.

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Physics for NEET 2027: Study Material, Mock Test, Chapters, Important Topics & More

"Physics" comes from the Greek word *"physike"*, meaning "knowledge of nature", and refers to the scientific study of matter, energy, motion, and force, and the fundamental laws that govern the behaviour of the universe. Sir Isaac Newton is regarded as one of the principal architects of modern Physics for his laws of motion and universal gravitation. Physics forms one of the three subjects tested in NEET, contributing 45 questions worth 180 marks out of the total 720 marks. Physics is important for students from Class 6 to Class 12 in the CBSE syllabus and forms the conceptual backbone for engineering and medical entrance exams alike.

Physics Definition

Physics is the branch of science that deals with the study of matter, energy, and the interaction between them. It explains natural phenomena through mathematical laws covering motion, force, heat, light, sound, electricity, magnetism, and the structure of atoms and nuclei. Physics is studied from the sub-atomic scale of quarks and electrons to the cosmic scale of galaxies and black holes.

What is Physics?

Physics is a branch of science involving the systematic study of the natural world through observation, experimentation, and mathematical modelling. In different aspects of physical science, physics covers mechanics, thermodynamics, electromagnetism, optics, and modern physics. The term “physics” originates from the Greek word for “nature”, reflecting its early focus on explaining natural phenomena. Modern physics has expanded to include quantum mechanics, relativity, and particle physics, driving advances in technology, medicine, and space exploration. Broadly, physics meaning is the scientific study of matter, energy, and the fundamental forces that shape the universe.

History of Physics

The study of physics traces its roots back to ancient civilisations, where early thinkers observed the motion of celestial bodies and the behaviour of everyday objects. Over time, these observations evolved into systematic scientific investigation. Ancient and Renaissance-era scholars laid the foundation of this branch of science through experimentation and mathematical description of natural laws.

Early Physics Pioneers

- **Galileo Galilei:** Often called the Father of Modern Physics, his experiments on motion and his support for heliocentrism laid the groundwork for the scientific method in physics.
- **Sir Isaac Newton:** His three laws of motion and the law of universal gravitation, published in *Principia Mathematica* (1687), unified terrestrial and celestial mechanics and remain central to the NEET Physics syllabus.

New Advances in the Science of Physics

The 20th century transformed physics with Albert Einstein's theory of relativity, which redefined our understanding of space, time, and gravity, and the development of quantum mechanics by scientists such as Max Planck, Niels Bohr, and Erwin Schrödinger, which explained the behaviour of matter and energy at the atomic and sub-atomic scale. These developments underpin the Modern Physics unit tested in NEET.

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Branches of Physics

Physics is a wide field comprising several specialised branches, each concentrated on some particular aspect of the physical world. Some of the main streams in physics are as follows:

Branch	**Description**
Mechanics	The study of motion, forces, and energy in physical bodies, covering kinematics, dynamics, and rotational motion.
Thermodynamics	Deals with heat, temperature, and the conversion of energy between different forms.
Electromagnetism	Examines electric and magnetic fields, their interaction, and their effects on charged particles.
Optics	The study of light, its propagation, reflection, refraction, and wave behaviour.
Modern Physics	Covers atomic structure, nuclear physics, and the dual nature of matter and radiation.
Quantum Mechanics	Studies the behaviour of matter and energy at the atomic and sub-atomic scale.
Astrophysics	Applies the laws of physics to understand celestial bodies, stars, and the universe at large.
Acoustics	The scientific study of sound, its production, propagation, and properties as a wave.

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Important Parts of Physics

Given below is the list of important parts of physics based on the general areas that make up a typical NEET Physics syllabus:

Mechanics

Mechanics explains the motion of objects under the action of forces, covering kinematics, Newton's laws, work, energy, power, and rotational motion. This unit forms the conceptual foundation for the rest of the Physics syllabus and carries substantial NEET weightage.

- Motion in a Straight Line & Motion in a Plane
- Laws of Motion
- Work, Energy and Power
- System of Particles and Rotational Motion
- Gravitation

Properties of Matter and Thermodynamics

This section explains the mechanical and thermal properties of solids, liquids, and gases, along with the laws governing heat transfer and energy conversion.

- Mechanical Properties of Solids & Mechanical Properties of Fluids
- Thermal Properties of Matter
- Thermodynamics
- Kinetic Theory

Oscillations and Waves

This part explains periodic motion, sound waves, and their properties – a recurring source of NEET numericals.

- Oscillations
- Waves

Electrostatics and Current Electricity

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Electrostatics and Current Electricity together form one of the highest-weightage areas of NEET Physics, covering electric charges, fields, potential, capacitance, and the flow of electric current.

- Electric Charges and Fields
- Electrostatic Potential and Capacitance
- Current Electricity

Magnetism and Electromagnetic Induction

This unit explains magnetic fields, the forces they exert on moving charges, and how changing magnetic fields induce electric currents.

- Moving Charges and Magnetism
- Magnetism and Matter
- Electromagnetic Induction
- Alternating Current
- Electromagnetic Waves

Optics

Optics covers the behaviour of light through reflection, refraction, and wave phenomena, and is a diagram-heavy, high-scoring section of NEET Physics.

- Ray Optics and Optical Instruments
- Wave Optics

Modern Physics

Modern Physics explains the structure of the atom, the nucleus, and the dual nature of matter and radiation – concepts that underpin technologies such as nuclear energy and semiconductor devices.

- Dual Nature of Radiation and Matter
- Atoms & Nuclei
- Semiconductor Electronics: Materials, Devices and Simple Circuits

Units, Measurements and Practical Physics

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This foundational section covers physical quantities, dimensional analysis, error analysis, and experimental skills that are tested both directly and indirectly across the NEET Physics paper.

- Units and Measurements

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Importance of Physics

The importance of physics extends well beyond the classroom, contributing significantly to medical technology, engineering, and everyday life.

What does Physics contribute to?

1. Medical Technology: Physics principles underpin diagnostic and therapeutic tools such as X-rays, MRI, ultrasound, and radiotherapy, making it directly relevant to a career in medicine.

2. Space Exploration: Physics enables the design and navigation of spacecraft, satellites, and planetary missions, including India's own space programme.

3. Energy and Sustainability: Understanding thermodynamics and electromagnetism supports the development of renewable energy technologies such as solar cells and wind turbines.

4. Electronics and Communication: Semiconductor physics underlies modern computing, smartphones, and telecommunications infrastructure.

5. Fundamental Understanding of Nature: Physics provides the mathematical and conceptual tools to explain natural phenomena, from falling objects to the expansion of the universe.

News on Physics

Here are some recent developments in physics-related research relevant to India:

- **LIGO-India Observatory Breaks Ground**
  In April 2026, the LIGO-India gravitational-wave observatory broke ground at Aundha in Maharashtra's Hingoli district, positioning India as a key node in the global network of gravitational-wave detectors alongside facilities in the US, Italy, and Japan.
- **National Quantum Mission Advances**
  India's National Quantum Mission continues to fund research and startups in quantum sensing, quantum computing, and precision navigation technologies, including gravimeters and atomic clocks with applications for ISRO's satellite navigation systems.
- **Quantum Sensors for Fundamental Physics**
  Researchers worldwide, including Indian institutes, are developing ultra-sensitive quantum sensors using ultracold atoms that could help detect dark matter and ancient gravitational waves, pushing the boundaries of experimental physics.

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- **ISRO’s Continued Space Missions**
  The Indian Space Research Organisation (ISRO) continues to apply core physics principles – mechanics, thermodynamics, and electromagnetism – in its ongoing satellite launch and interplanetary mission programmes.

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Physics for NEET 2027

Physics contributes 45 questions worth 180 marks out of the total 720 marks in NEET – the same weightage as Botany and Zoology individually. Unlike Biology, where questions are concentrated in a few high-yield chapters, Physics chapters in NEET are allotted fairly evenly, which means comprehensive preparation across the syllabus matters more than chapter-skipping. For NEET preparation, covering the syllabus from the NCERT textbooks and practising numericals is essential.

Note: As per the latest reports, NEET 2027 is expected to retain the 180-question, 720-mark format (+4 for a correct answer, –1 for an incorrect answer, 0 for unattempted) and may move to Computer-Based Test (CBT) mode instead of the traditional pen-and-paper format. This change has not yet been confirmed by NTA and will be updated once the official NEET 2027 information bulletin is released.

NEET Physics Preparation

For effective preparation of Physics for NEET 2027, understanding the NEET Physics syllabus is the first and most important step. The syllabus for NEET Exam 2027 covers the chapters of CBSE Class 11 and 12. Having command of the topics and concepts of Physics classes 11 and 12 becomes the top priority for aspiring students. Given below is the chapter-wise weightage of NEET Physics questions asked over the last five years:

Chapter name	2021	2022	2023	2024	2025	2026	Total
Current Electricity	4	4	4	4	3	1	20
Electrostatic Potential and Capacitance	3	5	1	4	2	3	18
Ray Optics and Optical Instruments	4	4	3	2	1	3	17
Gravitation	3	3	3	2	3	2	16
Semiconductor Electronics	3	3	3	3	2	3	17
Moving Charges and Magnetism	3	3	3	2	2	1	14
Electromagnetic Induction	2	1	2	1	1	5	12

Oscillations	2	3	2	4	2	1	14
System of Particles and Rotational Motion	2	0	2	2	1	3	10
Units and Measurements	3	3	2	3	1	1	13
Electric Charges and Fields	4	3	3	1	1	1	13
Alternating Current	2	3	3	2	2	0	12
Electromagnetic Waves	2	2	1	2	2	2	11
Nuclei	3	3	1	1	0	0	8
Thermodynamics	1	1	1	3	2	2	10
Work, Energy and Power	1	1	2	2	3	2	11
Dual Nature of Matter and Radiation	1	2	0	1	2	2	8

Source: Careers360 NEET Physics chapter-wise weightage analysis (2021–2026), based on the original NEET 2026 exam held on 3 May 2026. Figures denote the number of questions asked from each chapter in that year's NEET paper.

Across 2021–2026, Current Electricity, Electrostatic Potential and Capacitance, Ray Optics, Gravitation, and Semiconductor Electronics remain the most consistently high-scoring Physics chapters. Note the sharp 2026 spike in Electromagnetic Induction (5 questions) and System of Particles and Rotational Motion (3 questions) – both worth extra attention this cycle – while Current Electricity, Alternating Current, and Nuclei saw an unusually low count in 2026 compared to their historical average.

Note: NEET 2026 was held on 3 May 2026 and re-conducted as Re-NEET 2026 on 21 June 2026 following a paper-leak-related cancellation. The 2026 column above reflects the original NEET 2026 paper. Careers360's Re-NEET 2026 analysis reports that Electromagnetic Induction and Alternating Current, followed by Optics, Electrostatic Potential and Capacitance, and System of Particles and Rotational Motion, were the most heavily tested chapters in the re-exam – a broadly similar high-weightage cluster to the original paper.

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Class 11 Physics Chapters for NEET 2027

Given below is the list of important Physics chapters from the Class 11 syllabus for NEET Exam preparation:

Units and Measurements	Motion in a Straight Line
Motion in a Plane	Laws of Motion
Work, Energy and Power	System of Particles and Rotational Motion
Gravitation	Thermal Properties of Matter
Thermodynamics	Kinetic Theory
Oscillations	Waves

Also Read:

- NCERT solutions for class 11 Physics
- NCERT exemplar solutions for class 11 Physics

Class 12 Physics Chapters for NEET 2027

Given below is the list of important Physics chapters from the Class 12 syllabus for NEET Exam preparation:

Electric Charges and Fields	Electrostatic Potential and Capacitance
Current Electricity	Moving Charges and Magnetism
Magnetism and Matter	Electromagnetic Induction
Alternating Current	Electromagnetic Waves
Ray Optics and Optical Instruments	Wave Optics
Dual Nature of Radiation and Matter	Atoms
Nuclei	Semiconductor Electronics

Also Read:

- NCERT exemplar solutions for class 12 Physics
- NCERT solutions for class 12 Physics

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Tips for Physics Preparation

Here are some tips to prepare Physics for the NEET Exam:

- **Focus on the NCERT Textbook:** Study the NCERT textbook thoroughly – many NEET Physics questions are based directly on NCERT theory, formulas, and solved examples.
- **Build Strong Fundamentals:** Physics is cumulative; a weak grasp of Class 11 mechanics makes Class 12 electromagnetism harder to master.
- **Practice Numericals Daily:** Physics in NEET is heavily numerical – daily problem-solving practice is non-negotiable.
- **Make a Formula Book:** Maintain a consolidated formula sheet for quick last-minute revision across all chapters.
- **Solve Previous Papers:** Regularly solve NEET and CBSE previous year questions to understand recurring numerical patterns.

Also read: NEET Physics Previous Year Papers

Best Physics Books for the NEET Exam

The best Physics books for NEET preparation are:

- NCERT textbook Physics Class 11
- NCERT textbook Physics Class 12
- Concepts of Physics by H.C. Verma
- Objective Physics by D.C. Pandey
- Fundamentals of Physics by Halliday, Resnick and Walker
- MTG's NCERT at Your Fingertips – Physics

All these books provide students with huge collections of MCQs, numericals, and practice papers for improvement in Physics NEET preparation.

Physics MCQ for NEET

Practice is very important in Physics for the NEET exam, since it gives clarity on formulas, units, and problem-solving approaches. Regular practice of similar question types increases familiarity with

exam patterns and helps ensure accuracy and speed under time pressure. Proper revision and focused numerical practice regularly help students score well in the NEET exam.

Practice MCQ On NEET Physics	Download PDF

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Additional Resources

Students can also download the NEET Mock Test 2027 for all subjects from the links given below:

Title	Download PDF
NEET Mock Test Physics 2027 PDF	Download
NEET Mock Test Biology 2027 PDF	Download
NEET Mock Test Chemistry 2027 PDF	Download
NEET 2027 Free Full-Length Mock Test PDF	Download

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Other Useful Resources

Some useful resources for NEET 2027 preparation are listed below:

Physics for NEET 2027 – High-Scoring Concepts

Want to score higher in NEET Physics? Certain chapters like Current Electricity, Electrostatics, and Ray Optics carry consistent weight in the NEET exam. This eBook compiles high-scoring concepts, important chapters, and practice mock tests to sharpen your understanding.

» Download now and master Physics for NEET!

NEET Physics Chapter-wise Weightage – 5-Year Analysis

This detailed Careers360 analysis breaks down the highest-weightage NEET Physics chapters based on five years of previous papers, along with preparation tips for each unit.

» Read the full 5-year Physics weightage analysis!

Top 50 NEET Physics Numericals – Guarantee 120+ Marks

A carefully selected set of the 50 most important Physics numericals covering high-weightage chapters, with step-by-step solutions to help you target 120+ marks in the Physics section.

» Read the Top 50 Physics Numericals guide!

NEET Physics Sample Paper 2027 (Real Exam Simulation, 45 Questions with Solutions)

Preparing for NEET 2027 and want to strengthen your Physics score with real exam-level practice? This NEET Physics Sample Paper simulates the actual exam experience with 45 questions and detailed solutions.

» Download the NEET Physics Sample Paper PDF now!

Top 100 NCERT-Based Questions for NEET 2027 (Physics, Chemistry & Biology)

A specially curated set of most-expected questions across all three subjects, designed strictly on NCERT concepts and previous-year trends, with step-by-step explanations.

» Download the Top 100 NCERT-Based Questions PDF!

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NEET Formula Sheet 2027 – Physics & Chemistry

A free, chapter-wise formula sheet compiling all the important Physics and Chemistry formulas for NEET 2027 in one place – perfect for last-minute revision.

» Download here and revise with the NEET 2027 Formula Sheet!

NEET Important Formulas – Physics

This free PDF brings you all the essential formulas from Mechanics, Optics, Electricity, Magnetism, and Modern Physics to help you solve numerical problems faster and more accurately.

» Download here and master Physics formulas for NEET!

Top 20 Multi-Concept Questions for NEET Physics with Detailed Solutions

Carefully selected multi-concept Physics questions from Mechanics, Electricity, and Modern Physics, along with detailed step-by-step solutions to improve conceptual clarity and accuracy.

» Download now and boost your NEET Physics preparation!

NEET 2027 Magnetism & Matter PYQs (50 Questions) – Free Mock Test with Detailed Solutions

This mock test includes 50 important previous year questions to help you master this high-weightage Physics chapter, following the latest NEET exam pattern.

» Download now to practise smartly and boost your NEET Physics score!

Electromagnetic Waves MCQs for NEET (Concept-Based Practice PDF)

Quickly revise and practise key concepts like the EM spectrum, its properties, and applications based on previous-year question trends, with clear explanations for each answer.

» Download now to strengthen your Physics preparation!

NEET vs NCERT Syllabus 2027: Common Chapters, Deleted Topics & Study Map

This resource helps you clearly understand the common chapters, deleted topics, and important areas to focus on from NCERT for NEET 2027, including for Physics.

» Download it now and study smarter with a clear roadmap!

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NEET Class 11 Physics Important Diagrams and Numerical Set-ups

Diagrams and numerical set-ups are an essential part of NEET Physics preparation. This resource provides quick-revision diagrams for optics, mechanics, and circuit-based questions.

» Download now and revise NEET Physics diagrams effectively!

Most Scoring Concepts for NEET (As Per Latest NTA Syllabus)

This comprehensive guide highlights high-weightage topics in Biology, Physics, and Chemistry that can significantly boost your score, with clear explanations and expert tips.

» Download now and ace your NEET preparation!

NEET Ultimate 60-Day Study Plan – Crack NEET in 2 Months

This free study plan shows you how to prepare for NEET effectively in just 60 days, with a daily schedule, chapter priorities, and smart revision tips.

» Download now and follow the NEET 60-Day Study Plan!

NEET Previous 10 Years Question Papers with Detailed Solutions

This eBook covers the last 10 years of NEET questions across Biology, Physics, and Chemistry with detailed, step-by-step solutions – ideal for identifying recurring numerical patterns.

» Download now and master previous years' papers!

NEET Chemistry Cheat Sheet – Must-Know Formulas for Quick Revision

A crisp, easy-to-revise formula cheat sheet for Physical, Organic, and Inorganic Chemistry – your perfect last-minute revision guide.

» Download now and revise Chemistry with the NEET Cheat Sheet!

Botany for NEET 2027: High-Scoring Concepts, Chapters, Mock Tests & Preparation Guide

The companion eBook to this one – covering the Botany half of the NEET Biology syllabus in the same chapter-wise, high-scoring format.

» Download the Botany for NEET 2027 eBook!

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Zoology for NEET 2027: High-Scoring Concepts, Chapters, Mock Tests & Preparation Guide

The companion eBook covering the Zoology half of the NEET Biology syllabus – Human Physiology, Reproduction, Genetics and Evolution – in the same chapter-wise format.

» Download the Zoology for NEET 2027 eBook!

NEET Previous 5 Years Question Papers with Solutions (2021–2025)

One resource, all papers! This PDF compiles NEET question papers from the last five years, complete with detailed solutions – a must-have for serious aspirants.

» Download now and access NEET's previous 5-year solved papers!

NEET 2026 Free Mock Test (PDF) with Solutions

A full-length NEET mock test simulating the real exam experience, with detailed solutions to help you evaluate your preparation level before NEET 2027.

» Download now and attempt the NEET Free Mock Test!

NEET 2026 Answer Key with Solutions (All Codes)

Practising with an official-pattern question paper and answer key is one of the most effective ways to analyse your performance ahead of NEET 2027.

» Download now and practice with the NEET 2026 Answer Key!

All resources above are from the Careers360 Medicine Prep library and open directly on medicine.careers360.com, school.careers360.com, or www.careers360.com. Titles marked as "2027" editions will be refreshed automatically once NTA releases the official NEET 2027 bulletin.

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